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DIFFERENTIAL RENTS AND THE DISTRIBUTION OF EARNINGS¹

By MICHAEL SATTINGER

A. Introduction

THIS paper generalizes the theory of differential rents to the case where two co-operating factors are heterogeneous. This generalization yields a theory in which the distribution of labour earnings depends on the distribution of capital among jobs and on the reserve prices of capital and labour. The resulting theory is essentially an outgrowth of Jan Tinbergen's 'supply and demand' model of the distribution of earnings [3, 4, 5, 6].

David Ricardo's presentation of a differential rents model was based upon an agricultural economy in which land is heterogeneous, differing by fertility, and labour is homogeneous [1]. In the model developed here, the two factors of production are labour and capital, and both are assumed to be heterogeneous. Individuals providing labour differ in their productive ability. Capital is assumed to take the form of discrete units called machines, which differ in size, and only one individual can work with a machine. This last assumption has the effect of eliminating the intensive margin, i.e. no extra output is gained by adding more labour to a machine at which an individual is already working.

B. Assumptions

Consider a large economy in which the distributions of individuals and machines can be described by continuous density functions. Let k be the amount of capital a machine is equivalent to and suppose the values of k cover an interval. It is convenient to refer to k as the size of a machine. Let $K(k)$ be the density function of machines with respect to size, so that

$\int_{k_1}^{k_2} K(x) dx$ is the number of machines with size between k_1 and k_2 .

Similarly, let g be a measure of the productive ability of an individual and suppose the values of g cover an interval. Let $G(g)$ be the density function of individuals with respect to productive ability, so that $\int_{g_1}^{g_2} G(x) dx$ is the number of individuals with productive abilities between g_1 and g_2 .

Assume that the output obtained in the production process is homogeneous. Let $f(g, k)$ be the output produced by an individual of productive ability g using a machine of size k . It is assumed that the function f has

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continuous first and second order partial derivatives with respect to its arguments. Suppose further that the production function has positive marginal products and obeys the law of diminishing marginal returns.

The economy under consideration is characterized by perfect competition. In the context of the model, this means that no individual and no owner of capital has sufficient market power to affect the wage rates of individuals or the rental costs of machines.

C. Wage and rent functions

Let $w(g)$ be the wage function for individuals and let $r(k)$ be the rental cost function for machines. The objective of the following analysis is to derive these functions from the assumption of equilibrium, the production function $f(g, k)$ and the density functions $G(g)$ and $K(k)$.

The derivation proceeds in several steps. First, it is necessary to make a tentative assumption concerning the assignment of individuals to machines in equilibrium. This assumption is used to derive the implied wage and rental cost functions. Whether these functions are consistent with equilibrium depends on whether the second order conditions are satisfied. These conditions will be investigated later, and the class of production functions for which the tentative assumption is valid will be specified.

For purposes of deriving $w(g)$ and $r(k)$, it is therefore assumed that more productive individuals will work with larger machines. It follows that the n th individual, in order of decreasing productive ability, will be found working with the n th machine, in order of decreasing size. The number of machines with size greater than a given level k is $\int_k^{\bar{k}} K(x) dx$, where $\bar{k}$ is the largest size of a machine. Similarly, the number of individuals with productive ability above a given level g is $\int_g^{\bar{g}} G(x) dx$ where again $\bar{g}$ is the greatest productive ability. For k to be the size of machine used by the individual with productive ability g , the following equality must hold:

$$\int_k^{\bar{k}} K(x) dx = \int_g^{\bar{g}} G(x) dx.$$

Using this equality, it is possible to solve for the size of machine used by an individual with productive ability g in equilibrium, if the tentative assumption is valid. As long as the numbers of machines and workers are different, there will inevitably be some unemployment of one or the other. This unemployment will be discussed when reserve prices are introduced.

Unless there are intermediate intervals over which $G(g)$ or $K(k)$ are zero, the size of machine will be a strictly increasing function of productive ability. This relation may be written as $k(g)$ or, in its inverse form, as $g(k)$.

The next step is to derive expressions for the derivatives $r'(k)$ and $w'(g)$ from earnings and rent maximizing considerations. Consider the earnings that an individual with productive ability g^* could get in equilibrium. From his point of view the market rental costs for different sizes of machines can be taken as given by the function $r(k)$. The earnings for alternative levels of k are given by $f(g^*, k) - r(k)$. Earnings maximization requires that the derivative of earnings with respect to machine size be zero:

$$\frac{\partial f(g^*, k)}{\partial k} - r'(k) = 0 \quad \text{or} \quad r'(k) = \frac{\partial f(g^*, k)}{\partial k}. \quad (1)$$

This is a first order condition and does not hold for arbitrary values of k . It holds only for the value of k which maximizes the earnings of the individual with productive ability g^* . However, it does hold for all values of g^* and k such that $g^* = g(k)$. Thus the first order condition (1) can be used to derive the slope or derivative of the rent function for every size of machine. This derivative is

$$r'(k) = \left(\frac{\partial f(g^*, k)}{\partial k} \right)_{g^*=g(k)}. \quad (2)$$

Thus, to find the derivative of the rent function at k , one obtains the partial derivative of the production function with respect to its second argument (that is, one treats the first argument, g^* , as a constant and not as a function of k); this partial derivative is then evaluated at the value of g corresponding to k in equilibrium.

Similarly, consider the rent that an owner of a machine of size k^* can obtain using individuals with different productive abilities. This rent is $f(g, k) - w(g)$. Rent maximization requires that

$$w'(g) = \frac{\partial f(g, k^*)}{\partial g}. \quad (3)$$

Again (3) does not hold for arbitrary values of g but only for $g = g(k^*)$. As with the rent function, the derivative of the wage function is obtained by evaluating the partial derivative at the value of k which corresponds to g in equilibrium:

$$w'(g) = \left(\frac{\partial f(g, k^*)}{\partial g} \right)_{k^*=k(g)}. \quad (4)$$

The next step is to investigate whether the tentative assumption which yielded the equilibrium assignment given by $k(g)$ and which was used to derive (2) and (4) is valid. For this assumption to be valid, the second order conditions for earnings maximization by individuals and for rent maximization by owners of capital must be satisfied.

These second order conditions are

$$\frac{\partial^2 f(g^*, k)}{\partial k^2} - r''(k) \leq 0 \quad \text{for } k = k(g^*) \quad (5)$$

and
$$\frac{\partial^2 f(g, k^*)}{\partial g^2} - w''(g) \leq 0 \quad \text{for } g = g(k^*), \quad (6)$$

where $r''(k) = d^2 r / dk^2$ and $w''(g) = d^2 w / dg^2$.

We can now derive conditions under which the tentative assumption that more productive individuals work with larger machines is valid. From (2),

$$r''(k) = \left(\frac{\partial^2 f(g^*, k)}{\partial k^2} \right)_{g^*=g(k)} + \left(\frac{\partial^2 f(g^*, k)}{\partial g^* \partial k} \right)_{g^*=g(k)} \left(\frac{dg^*}{dk} \right).$$

Therefore

$$\left(\frac{\partial^2 f(g^*, k)}{\partial g^* \partial k} \right)_{g^*=g(k)} \left(\frac{dg^*}{dk} \right) = r''(k) - \left(\frac{\partial^2 f(g^*, k)}{\partial k^2} \right)_{g^*=g(k)} \geq 0,$$

because of the second order condition in (5).

It follows that if
$$\left(\frac{\partial^2 f(g^*, k)}{\partial g^* \partial k} \right)_{g^*=g(k)}$$

is positive, then dg^*/dk can only be positive in equilibrium, i.e. more productive individuals will work with larger machines. A sufficient (but not necessary) condition for this term to be positive is that $\partial^2 f / \partial g \partial k$ be positive for all values of g and k . This condition is stronger than the assumption of a declining marginal rate of technical substitution, which holds for movements along a given isoquant. The conclusion from the above discussion is that if $\partial^2 f / \partial g \partial k$ is positive for all values of g and k , then the tentative assumption is valid, i.e. more productive labourers work with larger machines in equilibrium.

The differential equations in (2) and (4), if solved, yield the machine rent function $r(k)$ and the wage function $w(g)$; these functions would, however, involve constants of integration. These constants are arbitrary in the sense that any values are consistent with the differential equations (2) and (4). In fact they are completely determined by the division of output between the marginally employed labour and the marginally employed machine, as will be discussed shortly. Further, if we know the value of one of the constants, we can immediately determine the value of the other from the condition that factor payments exhaust the product.

D. Reserve prices for labour and machines

The traditional way in which the output is divided among the marginally employed factors is to specify reserve prices, i.e. prices at which the owners of factors are indifferent between providing them for use in production and withholding them from the market. Let us suppose that the reserve price for labour is $\hat{w}$. An individual will agree to work as long as his earnings are at least $\hat{w}$, the same for all individuals. Similarly, let $\hat{r}$ be the reserve price for owners of capital (this reserve price may be zero).

Whenever individuals are unemployed, the wage of the least productive employed individual will be bid down to $\hat{w}$. Also, whenever machines are unemployed, the rental cost for the smallest machine used will be bid down to $\hat{r}$.

Several cases arise. Suppose there are more individuals than machines. Inevitably some individuals will be unemployed. If the production obtained from the smallest machine and the corresponding individual exceeds $\hat{w} + \hat{r}$, then the individual will receive the reserve price $\hat{w}$ as his wage, and the machine owner will receive the remaining amount of production. An analogous case arises if machines exceed individuals.

In the general case, production falls below $\hat{w} + \hat{r}$ before either machines or individuals run out. Both machines and individuals would then be unemployed. Let us now consider this last case in more detail.

Consider an individual of arbitrary productive ability g_0 , and suppose $k_0 = k(g_0)$ is the size of the machine he would work with in equilibrium, if he worked. If

$$f(g_0, k_0) < \hat{w} + \hat{r},$$

then neither the individual nor the machine would be employed, since either the individual or the machine owner would receive insufficient compensation. On the other hand, if

$$f(g_0, k_0) > \hat{w} + \hat{r}$$

then output is sufficient to cover the reserve prices of the two factors of production. Further, the factor payments $w(g_0)$ and $r(k_0)$ must equal or exceed the reserve prices. If, say, $r(k_0) < \hat{r}$, then the individual with productive ability g_0 would be willing to work for less than $f(g_0, k_0) - r(k_0)$ rather than not work. This would increase the return to the owner of capital until $r(k_0) \geq \hat{r}$.

The least productive individual employed is the one for whom $f(g, k(g)) = \hat{w} + \hat{r}$. Then $w(g) = \hat{w}$ and $r(k) = \hat{r}$. Any individual with a lower productive ability would not work and any machine with a lower size would not be used. Given the value of g for which $w(g) = \hat{w}$, the whole wage function can be determined from the differential equation (4). Similarly, given the value of k at which $r(k) = \hat{r}$, the whole rent function can be determined using (2).

Thus the reserve prices $\hat{w}$ and $\hat{r}$ determine the least productive individual and smallest machine employed, the division of output between the last individual and last machine employed, and the constants of integration in the solutions of the differential equations (2) and (4).

Once the wage function $w(g)$ is determined, the size distribution of earnings can be constructed using the individual density function $G(g)$. Because the wage will be a strictly increasing function of productive ability, the

inverse function $g(w)$ can be defined. The number of individuals with productive ability between g_1 and g_2 is $\int_{g_1}^{g_2} G(g) dg$. Now let us treat g as a function of w and perform a transformation of variables on the integral. Then $dg = g'(w) dw$ and $\int_{w(g_1)}^{w(g_2)} G(g(w))g'(w) dw$ is the number of individuals with wages between $w(g_1)$ and $w(g_2)$. Thus the density function of individuals with respect to earnings is given by $G(g(w))g'(w)$.

E. Examples

All of the foregoing analysis will now be illustrated for specific functions. The easiest functions to work with are Pareto distributions for the density functions and the Cobb–Douglas for production.

Suppose the density functions for individuals and machines take the following Pareto (exponential) forms:

$$G(g) = cg^{-\gamma}, \quad \text{where } \gamma > 2 \text{ and } g \geq 1, \text{ and} \\ K(k) = hk^{-\delta}, \quad \text{where } \delta > 2, \text{ and } k \geq 1.$$

Suppose the production function is given by the Cobb–Douglas form

$$f(g, k) = g^\alpha k^\beta, \quad \text{where } 0 < \alpha < 1 \text{ and } 0 < \beta < 1.$$

The mixed partial derivative $\partial^2 f / \partial g \partial k = \alpha \beta g^{\alpha-1} k^{\beta-1} > 0$, so that more productive individuals will work with larger machines in equilibrium. If k is the size of machine used in equilibrium by the individual with productive ability g , then

$$\int_k^\infty hx^{-\delta} dx = \int_g^\infty cx^{-\gamma} dx \quad \text{or} \quad \frac{h}{\delta-1} k^{1-\delta} = \frac{c}{\gamma-1} g^{1-\gamma}$$

$$\text{or} \quad k(g) = \phi^{1/(1-\delta)} g^{(1-\gamma)/(1-\delta)}, \quad \text{where } \phi = c(\delta-1)/h(\gamma-1).$$

The partial derivatives of the production function are

$$\frac{\partial f}{\partial g} = \alpha g^{\alpha-1} k^\beta \quad \text{and} \quad \frac{\partial f}{\partial k} = \beta g^\alpha k^{\beta-1}.$$

Corresponding to (2), we thus have $w'(g) = \alpha g^{\alpha-1} k^\beta$ where k is given by $k(g)$, or

$$w'(g) = \alpha \phi^{-\beta/(\delta-1)} g^{-1+\theta/(\delta-1)}, \quad (7)$$

where $\theta = \alpha(\delta-1) + \beta(\gamma-1)$. Integration yields

$$w(g) = \frac{\alpha \phi^{-\beta/(\delta-1)}}{\theta/(\delta-1)} g^{\theta/(\delta-1)} + C_1, \quad (8)$$

where C_1 is the constant of integration. Similarly,

$$r'(k) = \beta \phi^{-\alpha/(\gamma-1)} k^{-1+\theta/(\gamma-1)} \quad (9)$$

and

$$r(k) = \frac{\beta \phi^{-\alpha/(\gamma-1)}}{\theta/(\gamma-1)} k^{\theta/(\gamma-1)} + C_2 \quad (10)$$

where C_2 is the constant of integration.

If the parameters are such that both individuals and machines are unemployed, then the constants of integration can be calculated and are

$$C_1 = (\beta(\gamma-1)\hat{w} - \alpha(\delta-1)\hat{r})/\theta \quad \text{and} \quad C_2 = -C_1. \quad (11)$$

The corresponding values for the least productive individual and smallest machine employed are

$$g_0 = \phi^{\beta/\theta}(\hat{w} + \hat{r})^{(\delta-1)/\theta} \quad \text{and} \quad k_0 = \phi^{-\alpha/\theta}(\hat{w} + \hat{r})^{(\gamma-1)/\theta}. \quad (12)$$

Suppose instead that the calculated value of g_0 in (12) is less than one while k_0 is greater than one. Then all individuals will be employed so that the least productive individual employed will be the one for whom $g = 1$. The corresponding machine size would be $\alpha^{1/(1-\delta)}$. The rental cost for that machine size would be $\hat{r}$ (because of the presence of unemployed machines) and the wage rate for $g = 1$ would be the difference between the production obtained and $\hat{r}$. The constants of integration would then be determined by these factor payments and not by the expressions in (11). Other cases can be derived in a similar manner.

Whether or not wages are a convex, concave, or linear function of g depends on the exponent of g in (7). This exponent in turn can be related to the amounts of inequality in the distributions of productive abilities and machine sizes. In a similar way these amounts of inequality can also be related to the shape of the machine rent function.

The Pareto coefficients $\gamma-1$ and $\delta-1$ (from the cumulative form for the Pareto distribution) are measures of the amount of equality in the distributions. A larger Pareto coefficient is regarded as indicating a more equal distribution. Suppose now that the production function has constant returns to scale, so that $\alpha+\beta = 1$. Three cases arise.

1. Suppose productive abilities are more equally distributed than machine sizes, so that $\gamma-1 > \delta-1$. Then the exponent of g in (7) is positive and wages are a convex function of productive ability. That is, the slope of the wage function goes up as productive ability increases. Also, the exponent of k in (9) is negative, so the machine rent is a concave function of machine size. The slope of the rent function decreases as machine size increases.

2. Suppose productive abilities are less equally distributed than machine sizes, so that $\gamma-1 < \delta-1$. Then wages are a concave function of productive abilities and rents are a convex function of machine size.

3. Suppose productive abilities are just as equally distributed as machine sizes, so that $\gamma-1 = \delta-1$. Then wages are a linear function of productive ability and rents are a linear function of machine sizes.

The density function for the distribution of earnings can be derived as described in the previous section and is given by

$$G(g(w))g'(w) = \frac{c\phi^{(1-\gamma)\beta/\theta}}{\alpha} \left[\frac{(w-C_1)\theta}{\alpha(\delta-1)} \right]^{-1-(\gamma-1)(\delta-1)/\theta}. \quad (13)$$

In the general case where $C_1 \neq 0$, earnings will not have the Pareto distribution. Even if $C_1 = 0$, so that earnings follow the Pareto distribution, the exponent will not in general be the same as the exponent in the distribution of productive abilities.

In a previous article [2], this author demonstrated that a minimum requirement for the distribution of earnings to differ in shape from the distribution of productive abilities was that there be comparative advantage in the performance of different tasks by different individuals. In the example that has been developed above, comparative advantage is ruled out by the multiplicatively separable form of the production function $f(g, k)$. Yet in general, the distribution of earnings and productive abilities are different shapes. Hence, the distributions can differ for another reason besides comparative advantage: the distribution of co-operating factors can affect the shape of distribution of earnings.

The inequality in earnings can be investigated in two stages. First, the distribution of g can be compared with that of $w-C_1$ in (13), and then this transform of the wage can be compared with the wage itself.

From (13), the Pareto coefficient for the distribution of the transform $w-C_1$ is

$$\frac{\gamma-1}{\alpha+\beta(\gamma-1)/(\delta-1)}.$$

This is greater or less than the Pareto coefficient for productive abilities, $\gamma-1$, depending on whether the denominator is less or greater than one.

For a given level of equality in the distribution of productive abilities, as measured by the Pareto coefficient $\gamma-1$, the denominator is an increasing function of $1-\delta$, the Pareto coefficient for the distribution of machine sizes. Hence greater values of the exponents α and β in the production function and more inequality in the distribution of machine sizes (as measured by the Pareto coefficient) result in more inequality in the distribution of the values of the transform $w-C_1$.

Restricting attention to the case where $\alpha+\beta = 1$, i.e. where the production function has constant returns to scale, the denominator will be less or greater than one depending on whether $1-\gamma$ is less or greater than $1-\delta$. Thus the transform $w-C_1$ will be more (less) equally distributed than productive abilities if productive abilities are less (more) equally distributed than machine sizes.

Next, let us compare the equality in the distribution of $w-C_1$ with the equality in the distribution of w . The comparison depends solely on the

sign of C_1 . If C_1 is positive, wages will be more equally distributed than the transform $w - C_1$. In turn, the sign of C_1 depends on the labour and machine reserve prices.

The effect of a larger reserve price for labour (and a correspondingly lower reserve price for machines) is to add a lump-sum amount to the earnings of all employed individuals, thereby reducing their earnings inequality. The matter is more complicated if the reserve price for labour increases while the reserve price for machines remains the same. Then the wages of each employed individual are increased by the same amount (but not by as much as the increase in the labour reserve price); however, fewer individuals will be employed. The net effect on earnings inequality is ambiguous. An increase in the reserve price of machines is unambiguously to increase labour earnings inequality. Fewer individuals are employed, and an amount, less than the change in the reserve price for machines, is subtracted from each individual's earnings.

In summary, for a given amount of inequality in productive abilities, inequality in earnings will depend positively on

1. the exponents of the production function, α and β ;
2. the inequality in the distribution of machine sizes;
3. the reserve price for machines, $\hat{r}$;

and will depend negatively on

4. the reserve price for labour, $\hat{w}$, holding the sum $\hat{w} + \hat{r}$ constant.

The Pareto distributions and Cobb–Douglas production function that have been used above are particularly amenable to analysis; however, they do not realistically describe the distributions of individuals and machines. Let us therefore briefly consider a more complex economy.

Suppose that the distributions of individuals and machines are normal, truncated so that only positive values of g and k are included:

$$G(g) = \frac{P}{\sqrt{(2\pi)\sigma_1}} e^{-(g-\mu_1)^2/2\sigma_1^2}, \quad g > 0$$

and

$$K(k) = \frac{P}{\sqrt{(2\pi)\sigma_2}} e^{-(k-\mu_2)^2/2\sigma_2^2}, \quad k > 0.$$

Suppose further that the production function takes the constant elasticity of substitution form

$$f(g, k) = (\delta g^{-\rho} + (1-\delta)k^{-\rho})^{-1/\rho}.$$

Because this production function is not multiplicatively separable, there will arise comparative advantage. It can be shown that if ρ is positive, more productive individuals will have a comparative advantage working with larger machines. If ρ is negative, more productive individuals have a comparative advantage working with smaller machines. The borderline

case occurs when ρ is zero; this reduces to the Cobb–Douglas case, where there is no comparative advantage. Even if ρ is negative, however, more productive individuals may be assigned to larger machines in equilibrium. It can be shown by differentiation that as long as $\rho > -1$, $\partial^2 f / \partial g \partial k$ will be positive so that dg/dk must be positive.

Assuming more productive individuals work with larger machines, the machine size that will be used by an individual of productive ability g in equilibrium can be found by transforming both k and g to the standardized normal and setting them equal:

$$\frac{g - \mu_1}{\sigma_1} = \frac{k - \mu_2}{\sigma_2} \quad \text{or} \quad k = \mu_2 + \frac{\sigma_2}{\sigma_1} (g - \mu_1).$$

When this condition holds, the cumulative density functions will be equal. Substituting $k(g)$ into the partial derivative $\partial f / \partial g$ yields the differential equation for the wage function:

$$w'(g) = \delta \left[\delta + (1 - \delta) \left\{ \left(\mu_2 - \mu_1 \frac{\sigma_2}{\sigma_1} \right) g^{-1} + \frac{\sigma_2}{\sigma_1} \right\}^{-\rho} \right]^{-1/\rho}.$$

By investigating $w''(g)$, it is possible to establish that the wage function will be a convex (upward-curving) function of productive ability whenever $\mu_2 - \mu_1 \sigma_2 / \sigma_1$ is negative, i.e. whenever

$$\frac{\sigma_2}{\mu_2} > \frac{\sigma_1}{\mu_1}.$$

The ratio of the standard deviation to the mean of a distribution is called the coefficient of variation and is a good measure of the amount of inequality, since it does not depend on the units of measurement. The above result then states that the wage function will be convex whenever machine sizes are more unequally distributed than productive abilities. Similarly, the wage function will be concave whenever machine sizes are less unequally distributed than productive abilities. The borderline case occurs when productive abilities and machine sizes have the same amount of inequality; then wages will be a linear function of productive ability.

As in the previous example, the shape of the wage function can be related to the amount of earnings inequality. A convex wage function tends to raise the earnings of more productive individuals relative to less productive individuals, thereby increasing earnings inequality. However, the final result also depends upon the constant of integration.

It does not appear possible to solve the differential equations explicitly to get $w(g)$ and $r(k)$. However, numerical techniques, in which the differential equations are solved via small iterations, can be used to describe the resulting wage and machine rent functions.

F. Possible generalizations and conclusions

The development of the theory in the foregoing sections has proceeded under the assumption that only one individual could work with a piece of capital or machine. This assumption had the effect of eliminating the intensive margin. It was then possible to specify the assignment of machines to individuals, given by the function $k(g)$, without first knowing the wage rates and machine rental rates. When variable proportions of individuals to machines are possible, the optimal number of individuals to be combined with a machine depends on the factor prices. Hence the assignment must be determined simultaneously with the wage and rent functions. In principle the system can still be solved; numerical techniques can also be applied to obtain close approximations to the solution. However, the matter will not be pursued further here.

A second possible generalization would be to explain the evolution of the distribution of machine sizes. The model that has been developed is essentially static: it describes the temporary equilibrium that arises for given distributions of productive abilities and machine sizes. However, the dynamic elements are best introduced in an empirical version of the model.

Another generalization would be to introduce a third heterogeneous factor, education or human capital. One could then investigate the optimal distribution of educational levels and the returns to education as determined by the distribution of jobs according to capital intensity.

But this paper is long enough. A reader who is not convinced of the merit of the model that has been developed, even with its shortcomings, will not be persuaded by further extensions. The points that have been established within the confines of the model are as follows:

1. The process by which the distribution of earnings is determined is similar in nature to the way in which differential rents are determined.
2. In an economy in which two heterogeneous factors are combined to yield production, the distribution of one of the factors affects the distribution of returns to the other factor.
3. When both labour and capital are heterogeneous, the distribution of earnings will in general not be of the same shape as the distribution of productive abilities.
4. In particular, the more unequally machine sizes are distributed (i.e. the more unequally capital is distributed among jobs), the more unequal will the distribution of earnings tend to be.
5. The distribution of earnings and earnings inequality also depend in a predictable way on the reserve prices of labour and capital.

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